Thermodynamics-1

Thermal Equilibrium :

$T_1 \longrightarrow T_2$ $(T_1 > T_2)$

→ Heat will flow from higher temp. to lower temp, untill temp of both sides become equal & const.

i.e $(T_1 = T_2)$

This state is called Thermal Eqm

First law of Thermodynamics :

→ Any gaseous system is represented by three physical parameters.

→ (P, V, T)

Also called Thermodynamic Coordinates of a gas

→ Paths through which these physical parameters are changed for a gas, are called 'Thermodynamic Processes'.

State-1 $(P_1\ V_1\ T_1)$

↓ Thermodynamic Process

State-2 $(P_2\ V_2\ T_2)$

Every thermodynamic process consist of three physical quantities

- Heat supplied 'Q'
- Work done by Gas 'W'
- Change in int. energy of gas 'ΔU'

→ First law states that total energy in every thermodynamic process is constant.

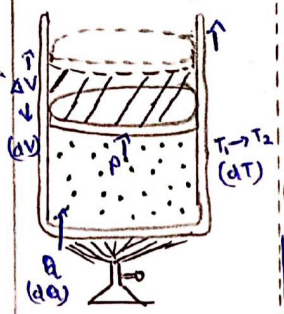
i.e

$$Q = W + \Delta U$$

$$dQ = dW + dU$$

Factors in a Thermodynamic Process :

→ Heat supplied to a gaseous sample, which ↑ses volume & temp. of the sample.



Some Pre-requisite

(i) Molar specific heat = c (J/mol-K)

(ii) It is of 2 type

(Const. vol.) $C_v = \frac{f}{2}R$

(Const press.) $C_p = \left(\frac{f+2}{2}\right)R$

(iii) Adiabatic constant (γ) is ratio of C_p & C_v and always > 1.

$$\gamma = \frac{C_p}{C_v} = \frac{f+2}{f}$$ and $\gamma > 1$ (Always)

(iv) Difference of C_p & C_v is always constant at a given temp.

$C_p - C_v = R$ (Mayer's Relation)

1) Heat supplied ⇒

$dQ = n \cdot c \cdot dT$ ['c' can be 'C_p' or 'C_v']

$\Delta Q = n \cdot c\,(T_2 - T_1)$

2) Change in Internal Energy ⇒

From KTG Knowledge, we know

$U = \frac{f}{2}nRT = nC_v \cdot T$

So,

$dU = n \cdot C_v \cdot dT$

$\Delta U = n \cdot C_v \cdot (T_2 - T_1)$

3) Work Done by Gas ⇒

↳ Final expression depends on the function of pressure for particular Thermodynamic Process.

$dW = P \cdot dV$

$$\Delta W = \int_{V_1}^{V_2} P \cdot dV$$

C_p & C_v for a Gaseous Mix. :

→ Consider a mixture of N-number of gases at const. temp. T

n_1	C_{v_1}	C_{p_1}
n_2	C_{v_2}	C_{p_2}
n_3	C_{v_3}	C_{p_3}
⋮	⋮	⋮

Then, For the mixture

$$C_{v_{eq}} = \frac{n_1 C_{v_1} + n_2 C_{v_2} + \cdots}{n_1 + n_2 + \cdots}$$

$$C_{p_{eq}} = \frac{n_1 C_{p_1} + n_2 C_{p_2} + \cdots}{n_1 + n_2 + \cdots}$$

$$\frac{n_1 + n_2 + \cdots}{\gamma_{eq} - 1} = \frac{n_1}{\gamma_1 - 1} + \frac{n_2}{\gamma_2 - 1} + \cdots$$

→ Other Emp relations ⇒

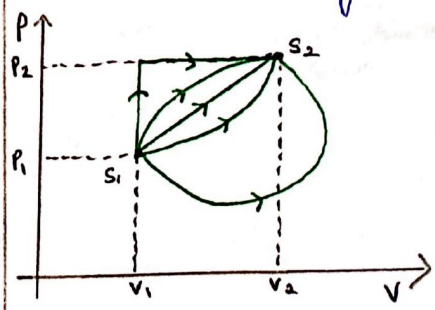
$C_v = \frac{R}{\gamma - 1}$ $C_p = \frac{\gamma R}{\gamma - 1}$

Indicator diagram of Thermodynamic Process :

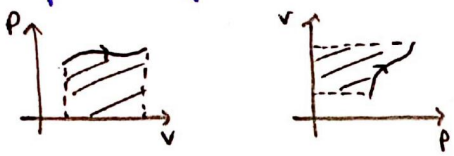
$(P_1\ V_1\ T_1)$ (S-1) $\xrightarrow{\text{Thermodynamic Process}}$ $(P_2\ V_2\ T_2)$ (S-2)

- Isochoric (V = const.)
- Isobaric (P = const.)
- Isothermal (T = const.) → $PV = nRT$
- Adiabatic (Q = 0).

→ Now, a gaseous sample can be taken from S-1 to S-2 through various processes, & each process has its own Indicator diagram (PV graph).



→ Area enclosed b/w PV graph gives magnitude of Work.



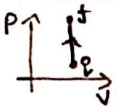
→ Work done by Gas ⇒ +ve ⇒ Expansion

Work done on Gas ⇒ −ve ⇒ Compression

→ Arrow on the curve is very imp., as it signifies whether the process is expansion or compression.

Standard Thermodynamic Process :

1) Isochoric Process ⇒ (V = Const.)



Since V = const., no change in vol. & no work is done.

→ By first law

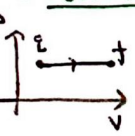
$\Delta U + \cancel{W}^{0} = Q$

$\boxed{\Delta U = Q}$

→ $\boxed{Q = n C_v (T_2 - T_1)}$

$\boxed{Q = \frac{nR}{\gamma - 1}(T_2 - T_1)}$

2) Isobaric Process ⇒ (P = const.)



(i) $\boxed{W = nR(T_2 - T_1) = P(V_2 - V_1)}$

(ii) $\boxed{Q = nC_p(T_2 - T_1) = \frac{n\gamma R}{\gamma - 1}(T_2 - T_1)}$

(iii) $\boxed{\Delta U = nC_v(T_2 - T_1) = \frac{nR}{\gamma - 1}(T_2 - T_1)}$

3) Isothermal Process ⇒ (T = const.)

Here, PV = const.

⇒ $P \propto \frac{1}{V}$ (Hyperbolic curve)

→ Also, by first law

$W + \cancel{\Delta U}^{0} = Q \Rightarrow \boxed{Q = W}$

(i) $\boxed{Q = W = nRT \ln\left(\frac{V_2}{V_1}\right)}$

$\boxed{Q = W = nRT \ln\left(\cancel{\frac{P_1}{P_2}}\right)\left(P_1/P_2\right)}$

4) Adiabatic Process ⇒ (Q = 0)

→ Graph is not exactly hyperbolic & more steeper.

$\boxed{\frac{dP}{dV} = -\gamma\frac{P}{V}}$ (Slope)

$\boxed{PV^{\gamma} = \text{const.}}$

→ Also,

$\boxed{P^{1-\gamma} \cdot T^{\gamma} = \text{const.}}$ $\boxed{T \cdot V^{\gamma - 1} = \text{const.}}$

→ By first law

$\cancel{Q}^{0} = W + \Delta U \Rightarrow \boxed{W = -\Delta U}$

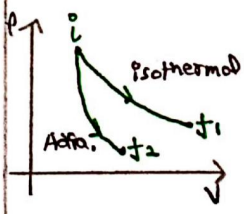
→ $\boxed{W = -\Delta U = -nC_v(T_2 - T_1)}$

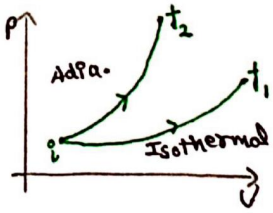
$\boxed{W = -\Delta U = \frac{nR}{\gamma - 1}(T_1 - T_2)}$

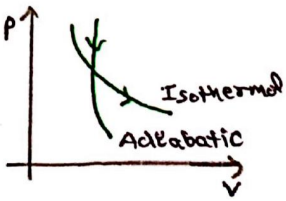
Slope of Isothermal & Adiabatic Curve :

$$\left|\frac{dP}{dV}\right|_{ad.} = \gamma \left|\frac{dP}{dV}\right|_{isoth.}$$

→ Since ($\gamma > 1$) always, so slope of adiabatic curve is always more than isothermal curve.







Reversible Polytropic Process :

→ Quasi-static processes are considered reversible process, where system is in eqm throughout the process & there is no energy loss (usually very slow process, terminating at ∞).

→ For polytropic processes

$\boxed{PV^{n} = \text{const.}}$

n → polytropic constant (not moles).

→ For such process, molar heat capacity is given by :

$\boxed{C = C_v + \frac{R}{1-n}}$

$\xrightarrow[n \to \infty]{V = \text{const.}} C = C_v$

$\xrightarrow[n \to 0]{P = \text{const.}} C = C_p$

$\xrightarrow{T = \text{const.}} C = \infty$

$\xrightarrow[n = \gamma]{Q = 0} C = C_v + \frac{R}{1-\gamma} = 0$

Also,

$\boxed{\Delta Q = n \cdot C \cdot \Delta T}$

$\boxed{\Delta W = \frac{nR(T_2 - T_1)}{1-n}}$

Eg: In a process, if $PV^2 = \text{const.}$, find heat req. to raise gas temp. from 300K to 500K (n = 2 mol, monoatomic gas).

Sol^n $\gamma = \frac{C_p}{C_v} = \frac{f+2}{f} = \frac{3+2}{3}$

$\boxed{\gamma = 5/3}$

Now,

$C = C_v + \frac{R}{1-n} \Rightarrow \frac{3R}{2} - R$

$\boxed{C = R/2}$

So,

$Q = n \cdot C \cdot (T_2 - T_1)$

$Q = 2 \times \frac{R}{2}(500 - 300)$

$\boxed{Q = 200R}$ Ans

NOTE :

(i) ΔU is state variable, and depends only on T_1 & T_2, & not on the path

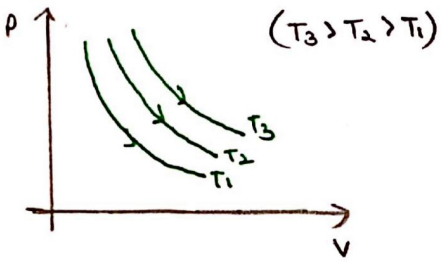
(ii) ΔQ & W, both are path depend as how P & V are related.

Thermodynamics -2

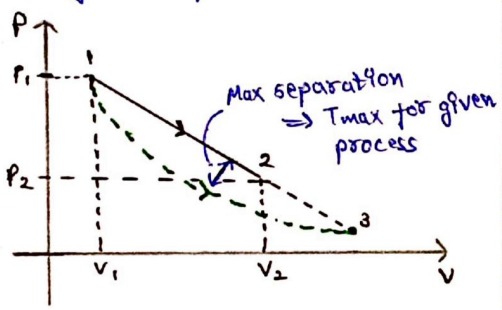
Temperature Variation in Linear Process:

→ Linear process is any process whose indicator diagram is a st. line. (generally with -ve slope)

→ For an isothermal process, graph moves upward ~~for~~ on increasing the temp.



→ Suppose we are given a following curve for the process.



→ If we compare given linear process with an isothermal process having same initial pt., then we observe

(i) First temp T ses & achives some max temp.

(ii) Then temp ↓ses gradually.

(iii) If the process is carried out further, then we will again reach at T_1 temp (point 3)

→ So, for above process, we can write relation b/w P & V as

$$\boxed{P = -mV + C_0}$$

By ideal gas eqn

$$T = \frac{PV}{nR} = \frac{P(-mV + C_0)V}{nR}$$

Put here

Put $\frac{dT}{dV} = 0$ & get $(V = _)$

→ Get value of T_{max}.

→ In case of +ve slope, simply treat process as a polytropic process of $n = -1$.

$$\boxed{PV^{-1} = \text{const}}$$

Free Expansion:

→ In this case, after removing clamps, gas expands w/o any opposition, so

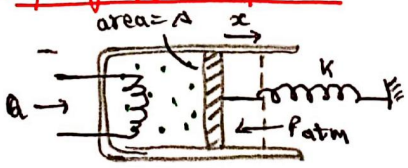
(i) Work done, $\boxed{W = 0}$

(ii) $\Delta Q = 0 \Rightarrow \Delta U = 0$

(iii) $\boxed{T = \text{const}} \Rightarrow T_1 = T_2$

(iv) $\boxed{P_1V_1 = P_2V_2}$

Spring Piston System:



Gas, $(P_1, V_1, T_1) \longrightarrow (P_2, V_2, T_2)$

By first law,

$$Q = \Delta U + W_{gas}$$

$$n \cdot C_v \cdot (T_2 - T_1) \qquad \left[P_{atm} \cdot \Delta V + \frac{1}{2}Kx^2\right]$$

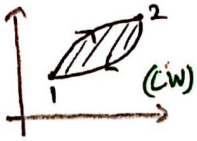
$(\Delta V = A \cdot x)$

Cyclic Processes:

→ In case of cyclic processes, area enclosed by indicator diagram cycle is work done.

→ $\boxed{\Delta U = 0}$ in any cyclic process.

↳ By first law, $\boxed{Q = W}$



⊙ W → +ve [Work done by gas]

⊙ W → −ve [Work done on gas]

⊙ Heat is +ve supplied

⊙ Heat is −ve rejected out

Second law of Thermodynamics:

Kelvin-Planck Statement ⇒

Impossible to convert 100% heat into work.

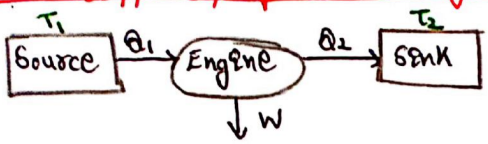
Heat Engines: Heat is supplied so that system does work.

Clausius Statement ⇒

Heat cannot flow from colder to hotter body without doing any external work.

Refrigerators: Work is done on the system to extract heat.

Thermal Efficiency of a Heat Engine:



$(T_1 > T_2)$

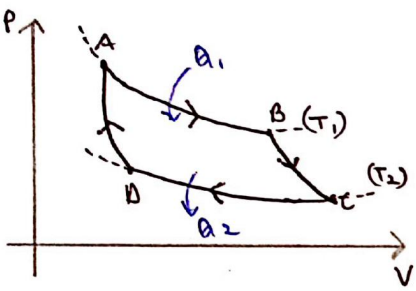
Some energy converted to work, rest wasted in sink.

1) $\boxed{W = Q_1 - Q_2}$ $(\because \Delta U = 0$, cyclic$)$

2) $\boxed{\eta = \frac{W}{Q_1} = 1 - \frac{Q_2}{Q_1}}$ (only magnitudes of Q_1 & Q_2)

(*) Carnot Engine ⇒

- Based on Carnot cycle (2 adiabatic + 2 isothermal)
- Yields max. possible η (Hypothetical).



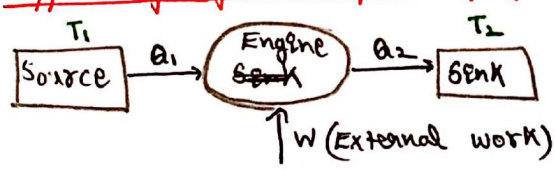
⊙ T is constant in A→B and C→D

⊙ $\Delta Q = 0$ in B→C and D→A

→ For a carnot engine

$\boxed{\frac{Q_1}{Q_2} = \frac{T_1}{T_2}}$ and $\boxed{\eta_{carnot} = 1 - \frac{T_2}{T_1}}$

Coefficient of Performance of a Refrigerator:



$(T_2 > T_1)$

1) $\boxed{Q_1 + W = Q_2}$

2) $\boxed{C.O.P = \frac{Q_1}{W} = \frac{Q_1}{Q_2 - Q_1}}$

3) If it follows carnot cycle i.e. $\frac{Q_1}{Q_2} = \frac{T_1}{T_2}$

$\Rightarrow$ $\boxed{C.O.P = \frac{T_1}{T_2 - T_1}}$

→ A best refrigeration cycle is one that removes the greatest amount of heat (Q_1) from inside the refrigerator for least amount of external work (W).

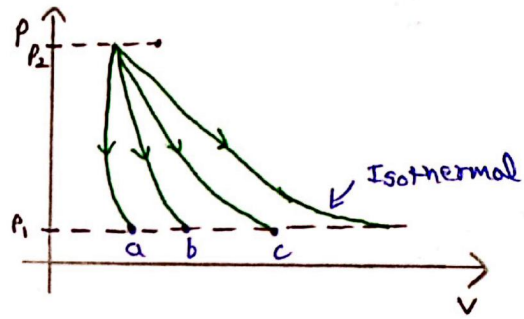
P-V slope among gases in adiabatic process:

→ since, $\left|\frac{dP}{dV}\right| = \gamma \frac{P}{V}$

a : monoatomic, $\gamma = 5/3 = 1.67$

b : diatomic, $\gamma = 7/5 = 1.4$

c : Polyatomic, $\gamma = 4/3 = 1.3$



NOTE: Every vibrational mode contributes to 2 DOF, one to KE & one for PE

i.e. If a gas sample has,

2 vib. mode → 4 DOF

10 vib. mode → 20 DOF